

Hannah Nguyen

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EDUCATION

University of California, Santa Barbara

Santa Barbara, California

Bachelor of Science in Statistics and Data Science and Applied Mathematics *Expected Graduation Date: June 2027*

- Relevant Coursework: Machine Learning, Probability and Statistics, Design and Analysis of Experiments, Regression Analysis, Theoretical Linear Algebra, Real Analysis, Numerical Analysis, Partial Differential Equations, Intermediate Python, Intermediate C++, SAS Base Programming
- Statistics and Data Science GPA: 3.78, Applied Math GPA: 3.82, Overall GPA: 3.81

EXPERIENCE

Data Scientist Intern

October 2024 – August 2025

IDXExchange LLC

Remote

- Collaborated on a prediction model as part of a team, then independently took over and continued development after team members exited early.
- Self-taught data cleaning, model training, and evaluation, while balancing coursework in statistics and data science.
- Analyzed large datasets and used problem-solving skills to improve efficiency and accuracy in project outcomes.
- Created a comprehensive guide to assist future interns, documented key methodologies and challenges.

CLAS Math Tutor

October 2025 – Present

University of California, Santa Barbara

Santa Barbara, California

- Provided one-on-one tutoring and drop in services to undergraduate students in lower division math 8 hours/week.
- Explained concepts, including algebra, calculus, and linear algebra, adjusted to fit different learning styles.
- Created a welcoming environment to encourage student engagement with the material and their classmates.
- Collaborated with mentors and supervisors to improve tutoring methods and support students effectively.

PROJECTS

Close Price Prediction Model | *Python, Numpy, Pandas, Git*

October 2024 - August 2025

- Developed a property closing price prediction model, using RandomForest, with a focus on hyperparameter tuning and model optimization to improve performance.
- Processed and cleaned numerical data (including outlier removal of top/bottom 1%) to improve data quality and model performance.
- Evaluated with multiple training/testing strategies, best model achieving MAE = 101k, RMSE = 243k, MAPE = 18.21%, $R^2 = 0.91$, on filtered test dataset.
- Documented progress and challenges in a detailed guide to assist future interns and ensure project continuity.

Interactive Calculator | *C++*

August 2024 – October 2024

- Built an interactive command-line calculator in C++ that parses and evaluates user-entered expressions.
- Implemented a memory system using a linked list storing up to 20 expressions, with options to view/delete entries.
- Added support for a “previous answer” variable, enabling users to reference their last result in new calculations.

Impact of Paperclip Positions on Final Distance | *R, ggplot2, RStudio*

May 2025 – June 2025

- Conducted a Factorial Design experiment testing the effects of paperclip positions on final distance.
- Performed sample size calculations and used linear modeling in R to conduct statistical analysis on data.
- Visualized and analyzed data using ggplot2 and kable tables.
- Practiced technical writing and clear communication, applying topic to broader context and referencing multiple peer reviewed articles, as well as interpreting results from analyses.

TECHNICAL SKILLS

Languages: Python (Intermediate), C++ (Intermediate), R (Intermediate), SAS Base Programming (Intermediate), LaTeX, HTML/CSS (Beginner)

Tools & Libraries: Python (pandas, NumPy, scikit-learn, PyTorch, Flask), R (ggplot2, dplyr), SQL, Excel

IDE/Text editor: Visual Studio Code, RStudio, Overleaf (for LaTeX)

Data Science: Probability and Statistics, Design of Experiments, Regression Analysis, Object-Oriented Programming